

Lecture 3: Lumped Elements

characteristics, and the LTI resistor

M. Mert Ankaralı

Department of Electrical and Electronics Engineering
Middle East Technical University

■ current ■ voltage ■ power ■ characteristic

The half we have not done

Network equations

KCL and KVL

Lecture 2

+

Terminal equations

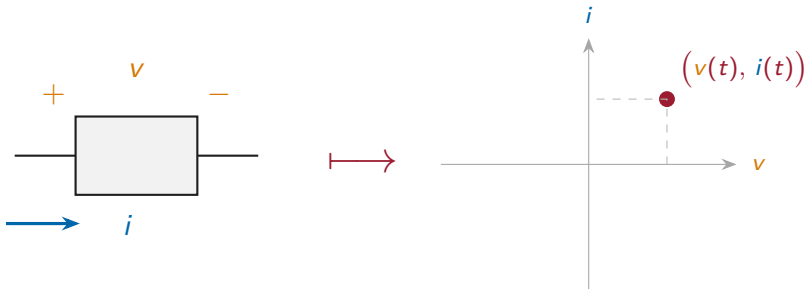
one per element

today

So far exactly one element has a terminal equation: $v = Ri$.

What can a two-terminal element be?

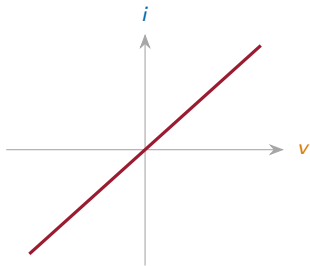
At every instant, one point in a plane



As time passes, the point moves.
The element is defined by **which points it is allowed to visit.**

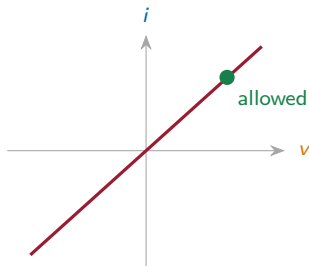
The characteristic

The **characteristic** of an element is the collection of **all** the pairs (v, i) it allows — a **curve** in the plane.



The characteristic

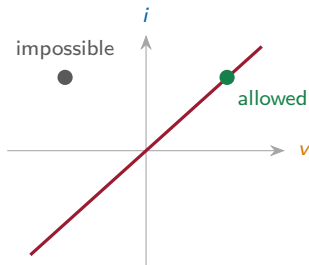
The **characteristic** of an element is the collection of **all** the pairs (v, i) it allows — a **curve** in the plane.



The operating point may sit **anywhere** on the **curve**

The characteristic

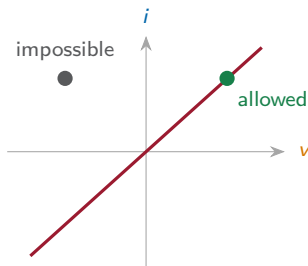
The **characteristic** of an element is the collection of **all** the pairs (v, i) it allows — a **curve** in the plane.



The operating point may sit **anywhere** on the **curve** and **nowhere** else.

The characteristic

The **characteristic** of an element is the collection of **all** the pairs (v, i) it allows — a **curve** in the plane.

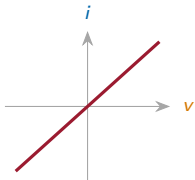


The operating point may sit **anywhere on the curve** and **nowhere else**.

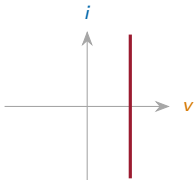
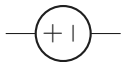
Give the element a voltage and the curve tells you the current — and the other way round.

Every element is a curve

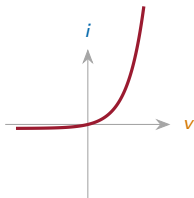
which curve?



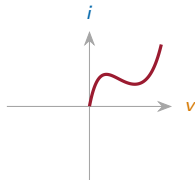
linear resistor



voltage source



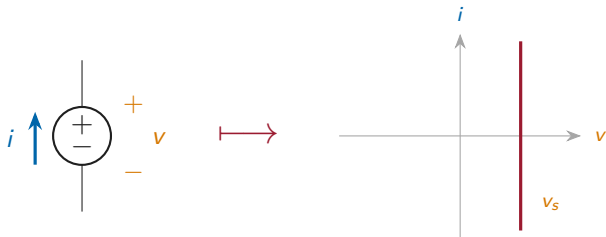
diode



tunnel diode

Four devices. Four curves. One plane.

The independent voltage source



$$v(t) = v_s(t)$$

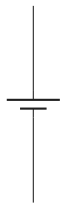
for every i

A **vertical line**, which may slide sideways as t runs.

Two symbols, one element



general source



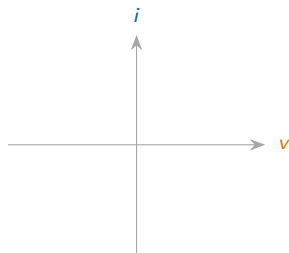
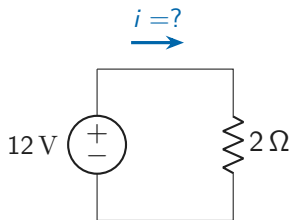
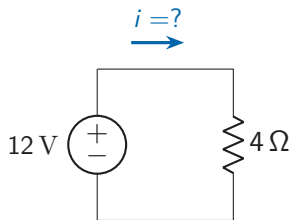
battery

Both are independent voltage sources; both appear in books, in the lab and in this course.

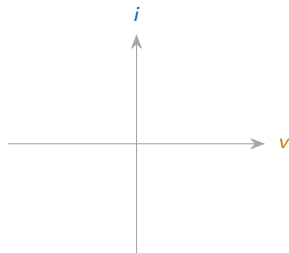
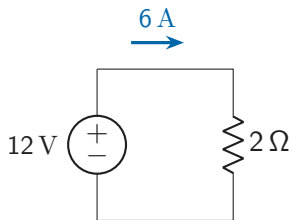
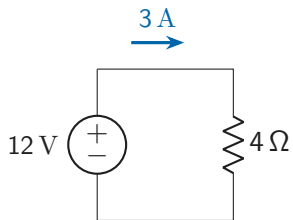
The battery symbol hints that the source is *constant*; the circle covers any $v_s(t)$.

We will use the **circle** from here on.

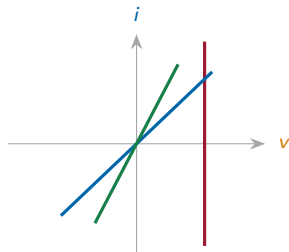
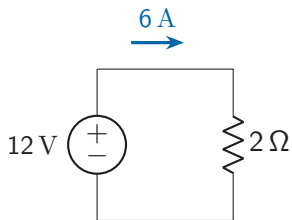
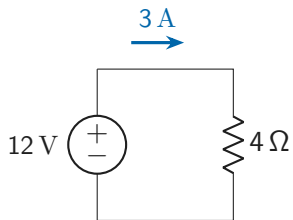
A source does not decide the other variable



A source does not decide the other variable

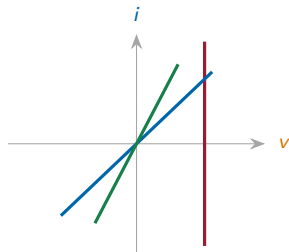
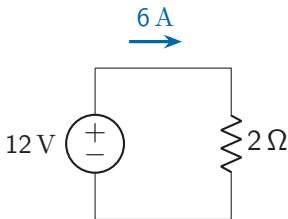
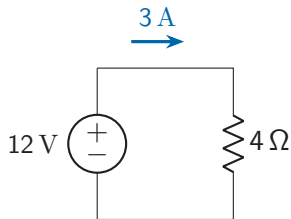


A source does not decide the other variable



The source's line **never moved**. The other element's line did.

A source does not decide the other variable



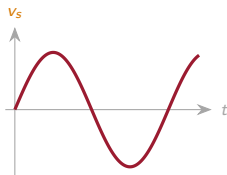
The source's line **never moved**. The other element's line did.

Which current flows is settled by KCL, KVL and the rest of the circuit — never by the source alone.

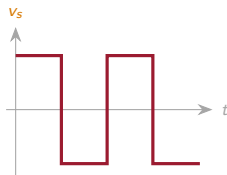
$v_s(t)$ can be any function you like



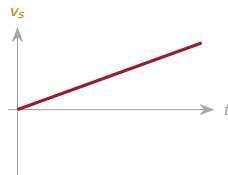
constant (DC)



sinusoidal



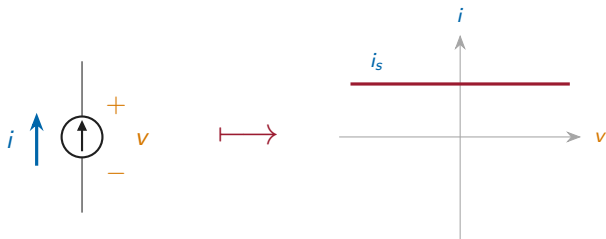
square



ramp

The characteristic stays a vertical line throughout — it just slides.

The independent current source



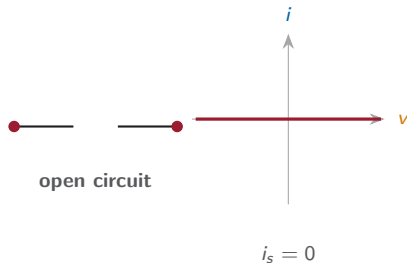
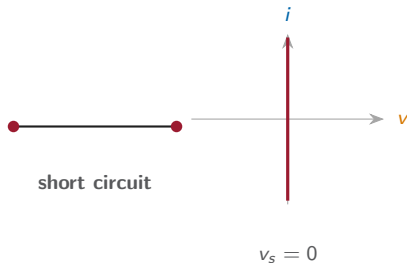
$$i(t) = i_s(t)$$

for every v

A **horizontal line** — everything on the last slides, with v and i exchanged.

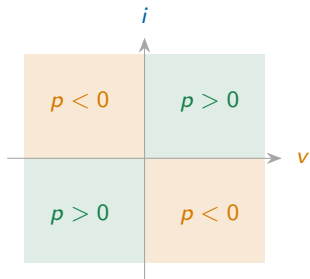
2 A into $4\ \Omega$ makes 8 V; into $100\ \Omega$ it makes 200 V; into an open circuit the ideal model demands infinity — which is the model warning you that a real source has a limit.

Two familiar elements, renamed



A short circuit *is* a voltage source of zero volts.
An open circuit *is* a current source of zero amperes.

Which quadrant is the element in?

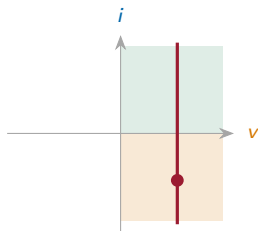


With the passive sign convention $p = vi$ is the power *absorbed*.

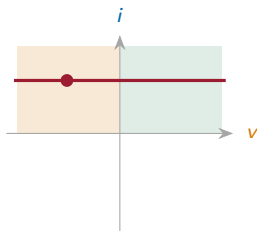
Stay in the **green** quadrants $\Rightarrow$ the element can only **absorb**: **passive**.

Reach an **orange** one $\Rightarrow$ it can **deliver**: **active**.

Every independent source is active



voltage source



current source

A vertical line at $v_s \neq 0$ crosses quadrants 1 *and* 4.

A horizontal line at $i_s \neq 0$ crosses quadrants 1 *and* 2.

Driving a circuit means giving out energy — so this had better be true.

What is a resistor?

A two-terminal element is a **resistor** if, at every instant, its characteristic is a **curve** in the $v-i$ plane.

Nothing about straight lines. Nothing about Ohm's law. Nothing about R .

The relation between v and i is **purely algebraic** — no derivatives, **no memory**.

So the family is large

Resistors

diodes, tunnel diodes, light bulbs, thermistors
short circuits, open circuits

independent sources — a straight line, just
not through the origin

Not resistors

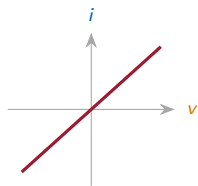
$$i = C \frac{dv}{dt} \quad v = L \frac{di}{dt}$$

the same v allows *any* i , depending on how v
is changing — so it is not a curve

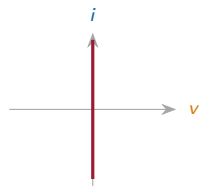
Yes — that makes a **battery** a resistor.

The word describes the *shape of the relation*, not dissipation. Read it as “**memoryless element**”.

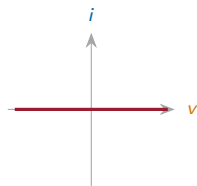
A gallery of resistors



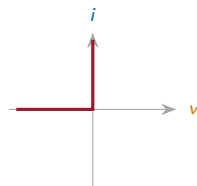
linear resistor



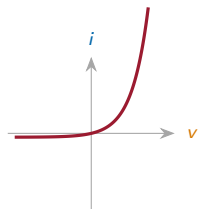
short circuit



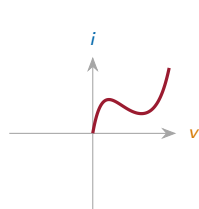
open circuit



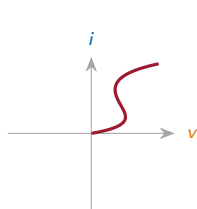
ideal diode



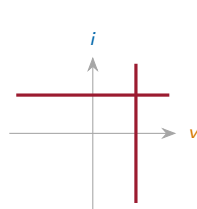
pn junction diode



tunnel diode



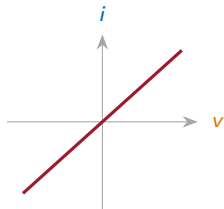
glow tube



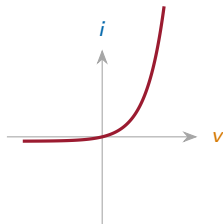
independent sources

Linear

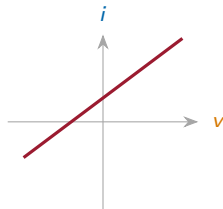
A straight line **through the origin**.



linear



bent: not linear

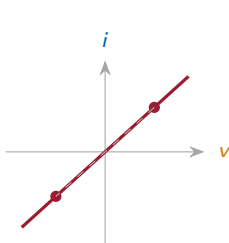


off the origin: not linear

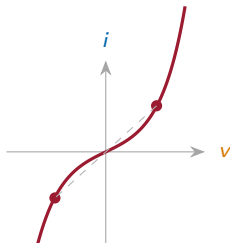
A battery is perfectly straight and **not** a linear element.

Bilateral

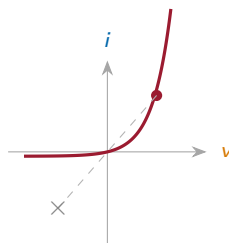
Symmetric about the origin: if (v, i) is on the curve, so is $(-v, -i)$.



bilateral



bilateral, *not* linear



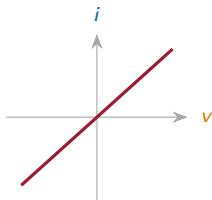
not bilateral

Turn the element round — does anything change?

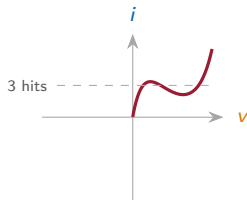
The middle one is two diodes back to back: **bent, yet symmetric**. Every linear element is bilateral; not the reverse.

Voltage-controlled or current-controlled?

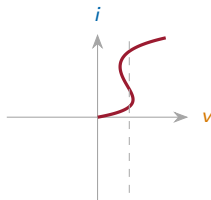
$i = f(v)$ or $v = g(i)$ — single-valued which way round?



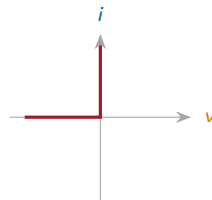
both



v -controlled only



i -controlled only

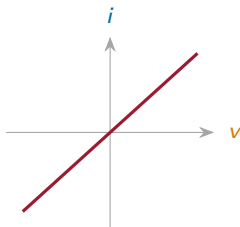


neither

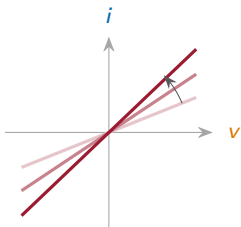
The ideal diode fails both: at $v = 0$ any positive current is allowed, and at $i = 0$ any negative voltage is.

Time-invariant

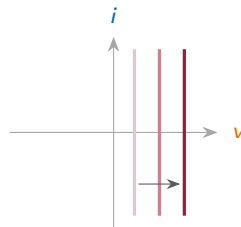
Is it **the same curve** at every instant?



time-invariant



a potentiometer being turned



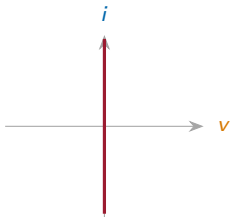
an AC wall socket

Still a curve at every instant — but a **different** one.

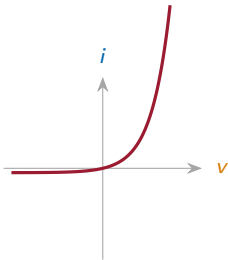
The gallery, characterized

Element	linear	bilateral	passive	v-contr.	i-contr.	time-inv.
linear resistor, $R > 0$	yes	yes	yes	yes	yes	yes
linear resistor, $R < 0$	yes	yes	no	yes	yes	yes
short circuit	yes	yes	yes	no	yes	yes
open circuit	yes	yes	yes	yes	no	yes
ideal diode	no	no	yes	no	no	yes
pn junction diode	no	no	yes	yes	yes	yes
tunnel diode	no	no	yes	yes	no	yes
glow tube	no	no	yes	no	yes	yes
DC voltage source, $v_s \neq 0$	no	no	no	no	yes	yes
DC current source, $i_s \neq 0$	no	no	no	yes	no	yes
AC voltage source	no	no	no	no	yes	no

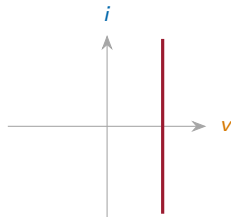
Three rows, read off the picture



short circuit

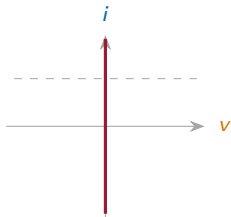


pn diode

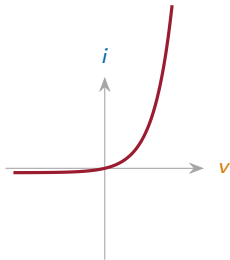


DC voltage source

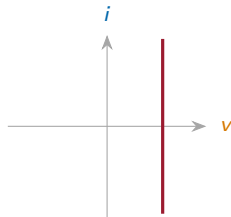
Three rows, read off the picture



short circuit



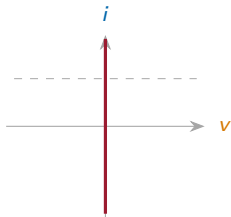
pn diode



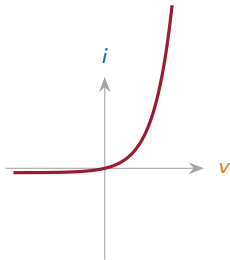
DC voltage source

At $v = 0$ any current is allowed $\Rightarrow$ not v -controlled.
But $v = g(i) = 0$ always $\Rightarrow$ i -controlled.

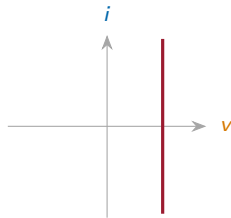
Three rows, read off the picture



short circuit



pn diode

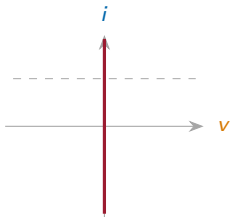


DC voltage source

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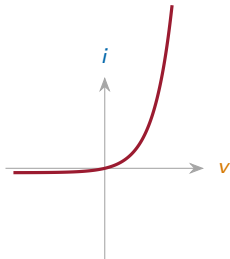
Single-valued and increasing both ways $\Rightarrow$ controlled by either. Stays in quadrants 1 and 3 $\Rightarrow$ passive.

Three rows, read off the picture



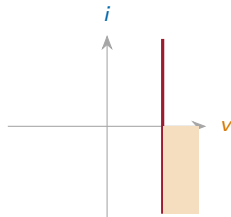
short circuit

At $v = 0$ any current is allowed $\Rightarrow$ not v -controlled. But $v = g(i) = 0$ always $\Rightarrow$ i -controlled.



pn diode

Single-valued and increasing both ways $\Rightarrow$ controlled by either. Stays in quadrants 1 and 3 $\Rightarrow$ passive.



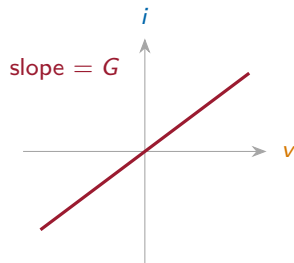
DC voltage source

Misses the origin $\Rightarrow$ not linear. Reaches quadrant 4 $\Rightarrow$ active. Any i at $v = v_s \Rightarrow$ not v -controlled.

The LTI resistor

Take two of those five words: **linear** and **time-invariant**.

A straight line through the origin, and it never moves. Only the slope is left.



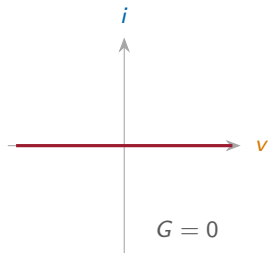
$$v = R i \quad \Longleftrightarrow \quad i = G v$$

$$G = 1/R$$

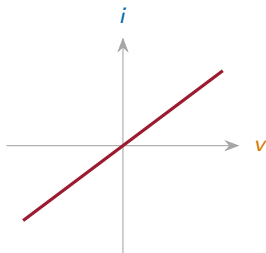
resistance R in ohms, $\Omega = \text{V/A}$

conductance G in siemens, $\text{S} = \text{A/V}$

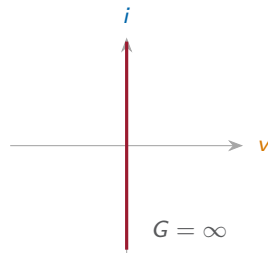
Turn the line and sweep R from ∞ to 0



$R = \infty$: open circuit



$0 < R < \infty$



$R = 0$: short circuit

A short circuit **fixes the voltage** at zero and lets any current through.
An open circuit **fixes the current** at zero and allows any voltage.

$G = 1/R$ — except at the two ends

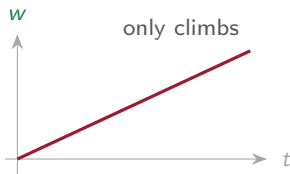
	R	G	characteristic
short circuit	0	∞	the i -axis, $v = 0$
open circuit	∞	0	the v -axis, $i = 0$

At each end only **one** of the pair is a usable number.

Same fact as Lecture 3: a short is current-controlled but not voltage-controlled, and an open circuit the other way round.

Power in an LTI resistor

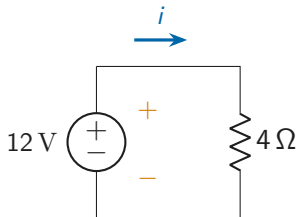
$$p = vi = Ri^2 = Gv^2 = \frac{v^2}{R}$$



Squares, so for $R > 0$ the power is **never negative**.

The energy graph can only climb: what goes in leaves as **heat**, at the same instant, and never comes back.

Example: a resistor as a heater



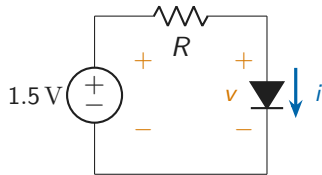
$$i = 3 \text{ A}$$

$$p = 12 \times 3 = 4(3)^2 = 36 \text{ W}$$

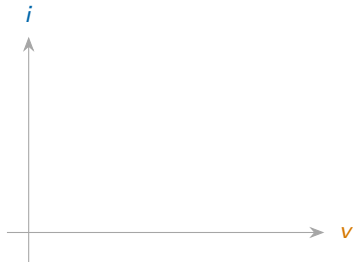
$$w|_{10 \text{ min}} = 36 \times 600 = 21.6 \text{ kJ}$$

All of it heat — which is why a resistor carries a **power rating** as well as a resistance.

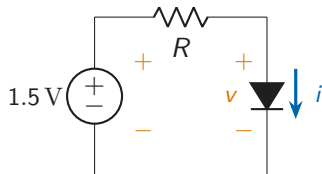
Example: a diode, a source and a resistor



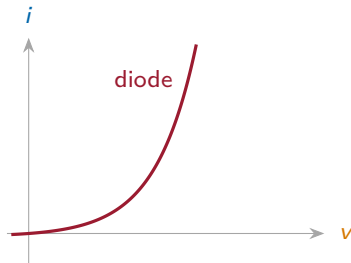
$v = ?$ $i = ?$



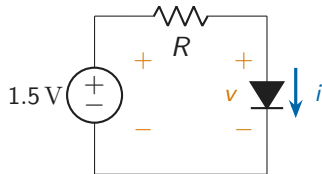
Example: a diode, a source and a resistor



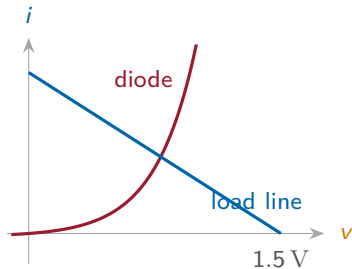
$$i = \frac{1.5 - v}{R}$$



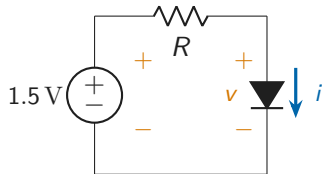
Example: a diode, a source and a resistor



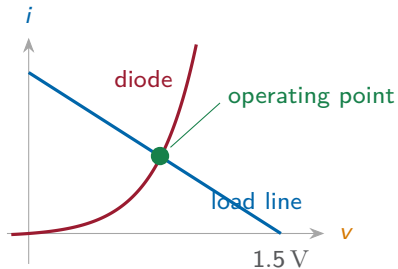
$$i = \frac{1.5 - v}{R}$$



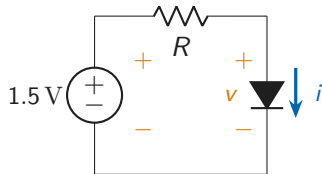
Example: a diode, a source and a resistor



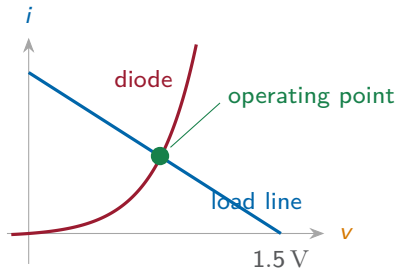
$$i = \frac{1.5 - v}{R}$$



Example: a diode, a source and a resistor



$$i = \frac{1.5 - v}{R}$$



Change R : the load line tilts about $(1.5 \text{ V}, 0)$. Change the source: it slides.
The diode's curve never moves — and **no algebra was needed**.

Summary

- An element is its **characteristic**: the set of (v, i) pairs it permits, drawn in the $v-i$ plane.
- **Voltage source**: vertical line, any current. **Current source**: horizontal line, any voltage. Neither decides the other variable.
- Short circuit = $v_s = 0$; open circuit = $i_s = 0$.
- A **resistor** is any element whose characteristic is a curve — algebraic, no memory. Diodes and sources are; capacitors and inductors are not.
- Five questions about the curve: **linear, bilateral, passive, v - or i -controlled, time-invariant**.
- **LTI resistor**: linear *and* time-invariant — one fixed line, $v = Ri$, $i = Gv$. $R = 0$ short, $R = \infty$ open. Its power is heat, at once, never returned.
- Element curve + **load line**, same plane: they cross where the circuit operates.

Next lecture

The capacitor

The first element that *remembers*. Its characteristic lives in a different plane altogether, and the energy you put into it comes back out.

Questions?

Notes for this lecture: see the course repository.